

Determinants of Outsourcing Purchase Prices: Regression Results and Data Notes

May 2026

Contents

1	Regression Results	2
1.1	T1: Firm-Level Variables	3
1.2	T2: Firm-Level Variables and Product Similarity	4
1.3	T3: Market-Level Variables Only	6
1.4	T4: Firm-Level and Market-Level Variables	7
1.5	T5: Full Specification	9
1.6	T6: Interactions — Firm Size $\times$ Market Conditions	12
2	SQL Data Extraction Logic	15
3	Bridge Data Source and Matching Rate	16

1 Regression Results

Overview

The dependent variable in all regressions is $\ln p_{fjct}$, the log outsourcing purchase unit price (CNY per unit) paid by firm f for product j (9-digit code) in city c in year t (2017). Three groups of explanatory variables are considered.

- **Firm-level variables (x):** $\ln \text{Output}_{ft}$ (log total VAT sales, size proxy); $\ln K_{ft}$ (log total assets, sourced from the tax-filing bridge, approximately 62.3% coverage; see Section 3); n_{ft}^{prod} (number of product codes with positive net output).
- **Market-level variables (z):** $\ln n_{jct}^B$ (log number of buyers of product j in city c , universe coverage); $\ln n_{jct}^S$ (log number of sellers, universe); $\ln q_{jct}^m$ (log total market purchase quantity, universe); $\ln \bar{p}_{jct}^m$ (log value-weighted market average price).
- **Product-similarity variables (s):** S_{mj} (`input_similarity`, cosine similarity of the input structure between the firm’s core product m and the outsourced product j); C_{mj} (`output_similarity`, output-structure similarity). Both vary at the firm $\times$ product level.

Sample. The sample consists of 3,410 firms randomly drawn from the VAT database (a time-saving compromise; will be expanded to the full sample in later work). The sample passes through the following filtering steps:

- **Buy-side filter:** 3,376 of the 3,410 firms have at least one valid 9-digit product code in the purchase records (after `bianma` filter).
- **Outsourcing definition:** an outsourcing product requires the *same* firm to appear on *both* the buy and sell side for the *same* product code. Only 2,108 firms satisfy this condition; these constitute `invoice_panel.dta`.
- **Intermediary removal:** 233 firms with outsourcing ratio $> 90\%$ (`is_intermediary == 1`) are dropped, leaving **1,875 firms** in the final regression panel (`reg_panel.dta`).

Tables T1, T2, T4, and T5 require $\ln K_{ft}$ and therefore have $N \approx 29,200$ (62.3% coverage); Tables T3 and T6 do not require capital data and have $N \approx 46,100$.

Product portfolio. Among the 3,376 buy-side firms, each purchases on average approximately 125 distinct product types (9-digit codes; $425,713 \text{ purchase records} \div 3,376 \approx 126$). Of these 3,376 firms, 2,108 (62.5%) have at least one product that appears on both their buy and sell sides (the outsourcing condition). These 2,108 outsourcing firms have on average 28.2 outsourcing product types (mean 28.2, median 3; the distribution

is highly right-skewed with a maximum of 925 products). In the final regression panel (`reg_panel.dta`, 1,875 firms after dropping intermediaries), the mean is 25.0 outsourcing products per firm (median 3).

Fixed effects and identification. All FE specifications include firm (δ_f), product (δ_j , 9-digit code), and/or city (δ_c , prefecture) fixed effects; standard errors are clustered at the firm level. Because the data cover a single cross-section (2017), firm-level variables ($\ln \text{Output}$, $\ln K$, n^{prod}) are perfectly collinear with the firm FE and are identified only in OLS specifications. Similarity variables (S_{mj} , C_{mj}) vary at the firm $\times$ product level and remain identified under firm FE. Market variables vary at the product $\times$ city level and remain identified under firm FE.

1.1 T1: Firm-Level Variables

The baseline equation is

$$\ln p_{fjct} = \gamma_1 \ln \text{Output}_{ft} + \gamma_2 \ln K_{ft} + \gamma_3 n_{ft}^{\text{prod}} + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \quad (1)$$

Table 1: Variable Definitions — Table T1

Variable	Definition
p_{fjct}	Dependent variable. Log outsourcing purchase unit price (CNY/unit), firm f , product j , city c , year t .
$\ln \text{Output}_{ft}$	Log annual VAT sales; proxy for firm size.
$\ln K_{ft}$	Log total assets from the tax-filing bridge ($\approx 62.3\%$ coverage). See Section 3.
n_{ft}^{prod}	Number of product codes with positive net output.
$\delta_f, \delta_j, \delta_c$	Firm, product (9-digit code), and city (prefecture) fixed effects.

Table 2: T1: Firm-Level Variables Only

	(1) OLS	(2) +Firm FE
ln Output	0.178*** (0.038)	—
ln K	−0.128*** (0.034)	—
n^{prod}	−0.001* (0.000)	—
Constant	3.273*** (0.469)	3.941*** (0.000)
Firm FE	No	Yes
Product FE	No	No
City FE	No	No
N	29,247	28,915
Adj. R^2	0.018	0.263

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. — = absorbed by fixed effects.

Firm-level variables are fully absorbed by Firm FE in single-year cross-section data. ln K is significantly negative in OLS, consistent with larger firms having stronger bargaining power.

1.2 T2: Firm-Level Variables and Product Similarity

The equation adds product-similarity terms:

$$\ln p_{fjct} = \gamma_1 \ln \text{Output}_{ft} + \gamma_2 \ln K_{ft} + \gamma_3 n_{ft}^{\text{prod}} + \beta_1 S_{mj} + \beta_2 C_{mj} + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \quad (2)$$

Table 3: Variable Definitions — Table T2 (additional variables)

Variable	Definition
S_{mj}	Input-structure similarity between the firm's core product m and outsourced product j (cosine similarity, $\in [0, 1]$). Varies at the firm $\times$ product level; identified under Firm FE.
C_{mj}	Output-structure similarity (overlap in downstream customers/ distribution channels). Same variation level as S_{mj} .
m	Core product: the product code with the highest net production value (= sales – same-product purchases) for firm f .

Table 4: T2: Firm-Level Variables and Product Similarity

	(1) OLS	(2) +Firm FE	(3) +Firm+Prod	(4) +Firm+Prod+City
ln Output	0.168*** (0.036)	—	—	—
ln K	−0.105*** (0.032)	—	—	—
n^{prod}	−0.001 (0.000)	—	—	—
S_{mj}	1.136*** (0.253)	0.341* (0.206)	0.080 (0.117)	0.080 (0.117)
C_{mj}	0.105 (0.304)	0.155 (0.219)	−0.093 (0.136)	−0.093 (0.137)
Constant	2.684*** (0.500)	3.853*** (0.032)	3.917*** (0.019)	3.917*** (0.020)
Firm FE	No	Yes	Yes	Yes
Product FE	No	No	Yes	Yes
City FE	No	No	No	Yes
N	29,237	28,905	28,552	28,552
Adj. R^2	0.028	0.264	0.589	0.586

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. — = absorbed by fixed effects.

S_{mj} is significantly positive in OLS (1.136***) and with Firm FE only (0.341*), but becomes insignificant once Product FE is included (0.080), indicating that the similarity effect is absorbed by product-level fixed heterogeneity.

1.3 T3: Market-Level Variables Only

The market-level specification is

$$\ln p_{fjct} = \lambda_1 \ln n_{jct}^B + \lambda_2 \ln n_{jct}^S + \lambda_3 \ln q_{jct}^m + \lambda_4 \ln \bar{p}_{jct}^m + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \quad (3)$$

Table 5: Variable Definitions — Table T3

Variable	Definition
$\ln n_{jct}^B$	Log number of buyers of product j in city c (universe coverage). Measures demand-side market thickness.
$\ln n_{jct}^S$	Log number of sellers of product j in city c (universe). Measures supply-side competition.
$\ln q_{jct}^m$	Log total market purchase quantity of product j in city c (universe).
$\ln \bar{p}_{jct}^m$	Log value-weighted market average price of product j in city c . Serves as an external price anchor.

Table 6: T3: Market-Level Variables Only

	(1) OLS	(2) +Firm FE	(3) +Firm+Prod	(4) +Firm+Prod+City
$\ln n^B$	0.154*** (0.048)	0.065** (0.029)	0.022 (0.036)	0.022 (0.036)
$\ln n^S$	0.051 (0.059)	0.038 (0.036)	−0.051 (0.034)	−0.051 (0.034)
$\ln q^m$	0.024 (0.015)	0.005 (0.013)	−0.003 (0.015)	−0.003 (0.015)
$\ln \bar{p}^m$	0.757*** (0.019)	0.624*** (0.015)	0.158*** (0.024)	0.158*** (0.024)
Constant	0.100 (0.239)	1.485*** (0.187)	3.792*** (0.267)	3.792*** (0.268)
Firm FE	No	Yes	Yes	Yes
Product FE	No	No	Yes	Yes
City FE	No	No	No	Yes
N	46,869	46,380	46,106	46,106
Adj. R^2	0.308	0.468	0.605	0.603

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

$\ln \bar{p}^m$ is robustly significant across all specifications (0.158*** to 0.757***). Buyer and seller counts become insignificant after Product FE is included.

1.4 T4: Firm-Level and Market-Level Variables

The combined firm and market specification is

$$\ln p_{fjct} = \boldsymbol{\gamma}'\mathbf{x}_{ft} + \boldsymbol{\lambda}'\mathbf{z}_{jct} + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \quad (4)$$

Table 7: T4: Firm-Level and Market-Level Variables

	(1) OLS	(2) +Firm FE	(3) +Firm+Prod	(4) +Firm+Prod+City
ln Output	0.135*** (0.032)	—	—	—
ln K	-0.085*** (0.026)	—	—	—
n^{prod}	-0.001*** (0.000)	—	—	—
ln n^B	0.108 (0.067)	0.098*** (0.035)	0.004 (0.049)	0.004 (0.049)
ln n^S	0.089 (0.079)	0.009 (0.047)	-0.048 (0.039)	-0.048 (0.040)
ln q^m	0.013 (0.019)	0.002 (0.017)	0.013 (0.019)	0.013 (0.019)
ln $\bar{p}^m$	0.732*** (0.022)	0.623*** (0.018)	0.181*** (0.033)	0.181*** (0.034)
Constant	-0.333 (0.451)	1.353*** (0.239)	3.393*** (0.378)	3.393*** (0.379)
Firm FE	No	Yes	Yes	Yes
Product FE	No	No	Yes	Yes
City FE	No	No	No	Yes
N	29,196	28,864	28,512	28,512
Adj. R^2	0.309	0.450	0.593	0.590

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. — = absorbed by fixed effects.

OLS identifies both firm-level and market-level variables jointly. Under FE, firm variables are absorbed by Firm FE; ln $\bar{p}^m$ remains robust throughout.

1.5 T5: Full Specification

The full specification combines all three variable groups:

$$\ln p_{fjct} = \boldsymbol{\gamma}'\mathbf{x}_{ft} + \boldsymbol{\beta}'\mathbf{s}_{fj} + \boldsymbol{\lambda}'\mathbf{z}_{jct} + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \quad (5)$$

Table 8: T5: Full Specification

	(1) OLS	(2) +Firm FE	(3) +Firm+Prod	(4) +Firm+Prod+City
$\ln \text{Output}$	0.131 ^{***} (0.031)	—	—	—
$\ln K$	-0.071 ^{***} (0.026)	—	—	—
n^{prod}	-0.001 ^{***} (0.000)	—	—	—
S_{mj}	0.769 ^{***} (0.192)	0.257 [*] (0.133)	0.106 (0.115)	0.106 (0.116)
C_{mj}	-0.016 (0.238)	0.023 (0.154)	-0.109 (0.135)	-0.109 (0.136)
$\ln n^B$	0.129 ^{**} (0.064)	0.098 ^{***} (0.035)	0.004 (0.049)	0.004 (0.049)
$\ln n^S$	0.075 (0.075)	0.012 (0.048)	-0.048 (0.039)	-0.048 (0.040)
$\ln q^m$	0.002 (0.019)	-0.001 (0.016)	0.013 (0.019)	0.013 (0.020)
$\ln \bar{p}^m$	0.717 ^{***} (0.022)	0.620 ^{***} (0.018)	0.181 ^{***} (0.033)	0.181 ^{***} (0.034)
Constant	-0.548 (0.446)	1.322 ^{***} (0.234)	3.391 ^{***} (0.376)	3.391 ^{***} (0.378)
Firm FE	No	Yes	Yes	Yes
Product FE	No	No	Yes	Yes
City FE	No	No	No	Yes
N	29,186	28,854	28,502	28,502
Adj. R^2	0.313	0.451	0.593	0.590

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. — = absorbed by fixed effects.

In full OLS, all three variable groups are jointly significant: $S_{mj} = 0.769^{***}$, $\ln K = -0.071^{***}$, $\ln \bar{p}^m = 0.717^{***}$. Under the full FE specification, only $\ln \bar{p}^m$ remains significant (0.181^{***}): firm variables are absorbed by Firm FE, and the similarity effect is absorbed by Product FE.

1.6 T6: Interactions — Firm Size $\times$ Market Conditions

To explore heterogeneous price-setting, all specifications include firm, product, and city FE, and the main effect of $\ln$ Output is absorbed by Firm FE. The equation is

$$\ln p_{fjct} = \boldsymbol{\lambda}' \mathbf{z}_{jct} + \sum_k \mu_k (\ln \text{Output}_{ft} \times z_{jct}^k) + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \quad (6)$$

The interaction terms vary at the firm $\times$ product level and remain identified under all three FE.

Table 9: Variable Definitions — Table T6 (interaction terms)

Variable	Definition
$\ln \text{Output} \times \ln n^B$	Size $\times$ buyer-market thickness. Expected sign: negative (larger firms benefit more from thick buyer markets).
$\ln \text{Output} \times \ln n^S$	Size $\times$ seller competition. Expected sign: negative.
$\ln \text{Output} \times \ln \bar{p}^m$	Size $\times$ market price level.
$\ln \text{Output} \times \ln q^m$	Size $\times$ market volume. Expected sign: negative.

Table 10: T6: Interactions — Firm Size $\times$ Market Conditions

	(1) Baseline	(2) +Size $\times$ nB	(3) +Size $\times$ nS	(4) +Size $\times\bar{p}^m$	(5) All
$\ln n^B$	0.022 (0.036)	0.413*** (0.101)	0.026 (0.036)	0.029 (0.036)	0.018 (0.137)
$\ln n^S$	-0.051 (0.034)	-0.057* (0.034)	0.415*** (0.115)	-0.053 (0.033)	0.132 (0.145)
$\ln q^m$	-0.003 (0.015)	-0.004 (0.015)	-0.005 (0.015)	-0.007 (0.015)	0.131* (0.070)
$\ln \bar{p}^m$	0.158*** (0.024)	0.157*** (0.024)	0.156*** (0.024)	-0.159** (0.080)	-0.015 (0.073)
Size $\times$ nB		-0.019*** (0.005)			0.001 (0.006)
Size $\times$ nS			-0.024*** (0.006)		-0.009 (0.007)
Size $\times\bar{p}^m$				0.016*** (0.004)	0.009** (0.004)
Size $\times$ qm					-0.007** (0.004)
Constant	3.792*** (0.268)	3.767*** (0.268)	3.785*** (0.265)	3.836*** (0.266)	3.811*** (0.264)
Firm FE	Yes	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes
N	46,106	46,106	46,106	46,106	46,106
Adj. R^2	0.603	0.603	0.603	0.604	0.604

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The main effect of $\ln \text{Output}$ is absorbed by Firm FE in all columns.

Individual interaction terms are significant when entered separately: larger firms pay lower prices in markets with more buyers ($\hat{\mu}_1 = -0.019^{***}$) and more sellers ($\hat{\mu}_2 = -0.024^{***}$). The size-market-price interaction is positive ($\hat{\mu}_3 = +0.016^{***}$). In the full specification (column 5), only $\hat{\mu}_3$ and $\hat{\mu}_4$ remain individually significant, likely due to multicollinearity among the interaction terms.

2 SQL Data Extraction Logic

Background

Raw data originate from the national VAT invoice database for 2017, stored in a relational database. Five aggregated views were extracted and saved as flat files for downstream Python processing. Table 11 summarizes each view.

Table 11: Extracted Data Views from the VAT Database

View name	Unit of observation	Raw / Clean rows	Contents
<code>firm_buy</code>	Buyer firm $\times$ product code	473,268 / 425,713	Purchase amount and quantity; 2,731 distinct products.
<code>firm_sell</code>	Seller firm $\times$ product code	101,844 / 89,584	Sales amount and quantity; 2,542 distinct products.
<code>firm_city</code>	Firm ID $\times$ prefecture city	3,410 / 3,410	City assignment for each sample firm. Used to merge city into purchase/sale files which contain no city field.
<code>city_buy</code>	Buyer prefecture $\times$ product code	2,104,901 / 1,393,329	Universe-level buyer count, amount, and quantity. Used to construct n^B , q^m , $\bar{p}^m$.
<code>city_sell</code>	Seller prefecture $\times$ product code	1,540,065 / 954,222	Universe-level seller count. Used to construct n^S .

“Universe” means *all* firms in the national VAT database, not only the 3,410 sample firms.

Core Python Processing Steps

1. **Product code standardization.** Raw codes are right-padded to 19 digits; the first 9 digits are retained as the standardized product code. Only the 2,778 valid codes from the national product catalog are kept.
2. **Outsourcing identification.** A firm-product pair is classified as an outsourcing transaction if and only if the firm appears in *both* `firm_buy` and `firm_sell` for the same product code (i.e., the firm both purchases and sells the same product), with $\min(\text{purchase quantity, sales quantity}) > 0$.

3. **City merge.** Because the purchase and sale files contain no geographic identifier, `firm_city` is joined to both files on firm ID to assign each transaction to a prefecture.
4. **Market condition construction.** Universe views `city_buy` and `city_sell` are aggregated at the product $\times$ city level to produce n_{jct}^B , n_{jct}^S , q_{jct}^m , and $\bar{p}_{jct}^m$. These aggregates are *not* restricted to sample firms, ensuring that the market-thickness measures capture the full competitive environment.
5. **Intermediary removal.** Firms flagged as intermediaries (`is_intermediary == 1`; outsourcing ratio $> 90\%$) are dropped from the 2,108-firm outsourcing panel. This removes 233 firms, leaving **1,875 firms** in the final regression panel (`reg_panel.dta`).

Sample Summary

Table 12: Sample Summary Statistics

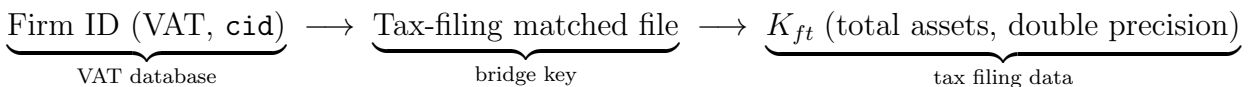
Statistic	Value
Original sample firms	3,410
Firms with valid buy-side product codes (<code>bianma</code>)	3,376
Outsourcing firms (<code>invoice_panel.dta</code>) (outsourcing = same product bought & sold)	2,108
Regression panel firms (<code>reg_panel.dta</code>) (after dropping 233 intermediaries)	1,875
Avg. outsourcing products per firm (mean / median)	25.0 / 3
Outsourcing product types (across all firms)	2,143
Regression observations (T3/T6)	$\approx 46,100$
Cities covered	262

3 Bridge Data Source and Matching Rate

Purpose

The VAT invoice database contains transaction-level information but no firm financial data. To include $\ln K_{ft}$ (log total assets) as a regressor in Tables T1, T2, T4, and T5, firm identifiers are bridged to a separate tax-filing dataset that records balance-sheet information.

Bridge Path



Matching Rate

Table 13: Bridge Matching Rate

Item	Value
Regression panel firms (post intermediary removal)	1,875
Successfully matched firms (with $\ln K$)	$\approx 1,168$
Firm-level matching rate	62.3%
Regression observations: T1/T2/T4/T5 (require $\ln K$)	$\approx 29,200$
Regression observations: T3/T6 (no $\ln K$)	$\approx 46,100$
Observation loss from capital requirement	$\approx 37\%$

The matching rate of 62.3% is substantially improved from the initial version ($\approx 48\%$) following an update to the bridge key.

Unmatched firms are dropped automatically by Stata when $\ln K$ is included as a regressor; Tables T3 and T6 are unaffected because they do not include capital.

$\ln K$ is absorbed by Firm FE in all FE specifications; its economic content derives solely from cross-sectional variation in the OLS columns.

Asset values are denominated in CNY (double precision); the log transformation is applied directly without rescaling.